

Classical Statistical Forecasting Baseline Methods:

1. Synthetic Dataset Creation

- **Components:**
 1. **Trend** → an upward slope over time.
 2. **Seasonality** → repeating yearly pattern (period = 365 days).
 3. **Noise** → random variation.
 - Data is split into:
 1. **Training set:** first 1,000 points.
 2. **Validation set:** remaining points.
 - The validation set preserves the trend and seasonality but zooms in on a smaller time window.
-

2. Naïve Forecast (Baseline)

Idea:

Predict the next value as *exactly* the last observed value.

Formula:

$$y^{t+1} = y^t$$

Performance:

- **MSE** ≈ 61.8
 - **MAE** ≈ 5.9
 - This is the baseline for comparison.
-

3. Moving Average Forecast

Idea:

Predict each point as the average of the last NN points (here N=30).

Effect:

- Smooths noise, but lags behind real changes (especially if seasonality exists).
 - **Result:** Worse error than naïve forecast because it doesn't account for trend/seasonality.
-

4. Seasonality Adjustment

Since the seasonality period = 365:

1. Compute **differenced series:**

$$y^t' = y^t - y^{t-365}$$

→ Removes repeating yearly pattern.

2. Apply moving average to y^t' (noise reduced, no seasonal swings).
-

3. Add **back the past seasonal values**:

$$\hat{y}_t = MA(y_{t'}) + y_t - 365$$

→ Produces predictions that follow trend + seasonality with less lag.

5. Double Moving Average Trick

- Instead of adding the **raw** seasonal values back, also smooth those seasonal values first.
- This further reduces noise, giving cleaner predictions.

Result:

- **Visually:** Prediction curve is smoother and closer to actual values.
 - **Numerically:** Lower error than all previous methods.
-

➤ Key Takeaways

- **Baseline first** → Always start with something simple (naïve forecast).
 - **Moving averages** can reduce noise but may hurt accuracy if not adjusted for seasonality.
 - **Seasonality differencing** is a simple but powerful way to improve forecasts.
 - Combining **trend + seasonality adjustments + smoothing** can yield strong classical forecasts before even touching machine learning.
-